1 2 3 5 7 9

# Bekat

Namangan avtobuslari uchun aqlli marshrut xaritasi



**Skanerlang**
Android — ilovani yuklab oling, iPhone — brauzerda oching
zedrubilan.github.io/bekat


**Namangan davlat texnika universiteti**
“Eng yaxshi kreativ g‘oya” respublika tanlovi · 2026
Mualliflar: Qodirov Ulug‘bek, Zamalitdinov Salim · 37-ATT-23 (r)


## 1 MUAMMO

### “Qaysi avtobus? Qayerda almashaman?”

- Avtobuslar ko‘paymoqda, yo‘nalishlar o‘zgarmoqda
- To‘g‘ri avtobus bo‘lmasa — odamlar taksiga o‘tadi
- Shaharlararo jadval faqat Telegram guruhlarida

**Chorsudan Yuksalishga qanday boraman?**

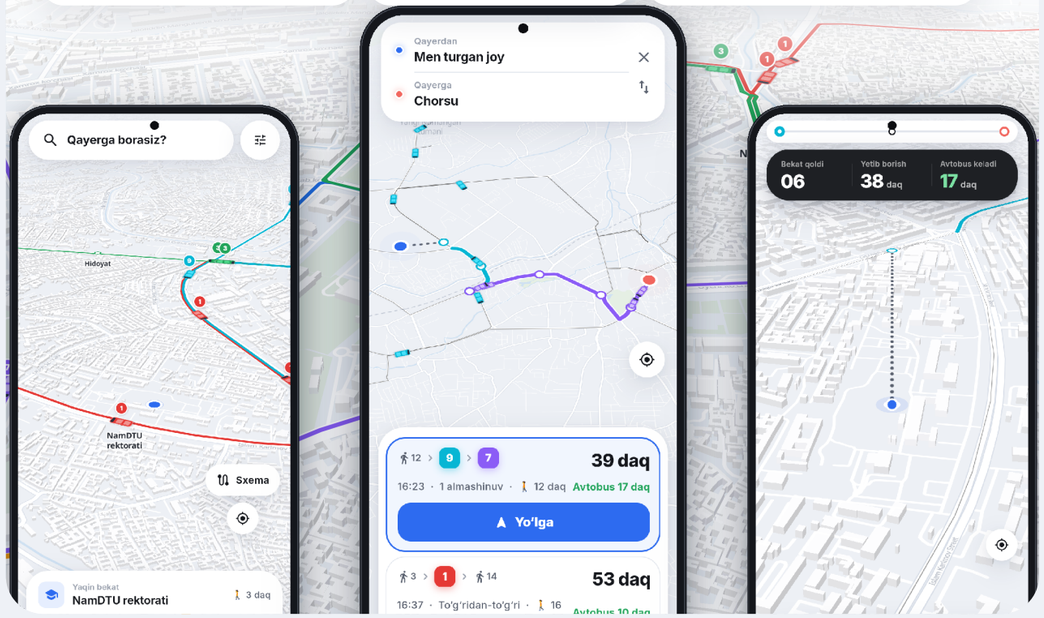
| | |
|---|---|
| **731,3k** Namangan shahri aholisi | **130** zamonaviy avtobus |
| **10** brutto-shartnomali yo‘nalish | **0** yo‘nalish ochiq xaritada (OSM) |

*Manbalar: namstat.uz (01.04.2026), yuz.uz (03.10.2026), gazeta.uz (24.09.2026), OpenStreetMap so‘rovi (14.10.2026)*

## 2 YECHIM

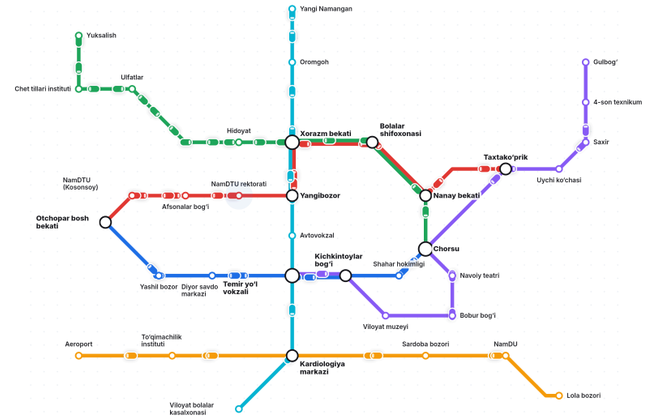
### Manzilni tanlang — avtobus, almashinuv va piyoda yo‘lni ko‘ring

№9 → №7 · 1 almashinuv · Avtobus 3 daqiqada · O‘zbek · Русский · English




## 3 METRO USLUBIDAGI SXEMA

### Butun tarmoq — bitta sxemada




1 Otchopar – Taxtako‘prik · tasdiqlangan
2 Otchopar – Chorsu · tasdiqlangan
3 Yuksalish – Chorsu · tasdiqlangan
5 Aeroport – Lola bozori · namuna
7 Gulbog‘ – Vokzal · namuna
9 Yangi Namangan – Viloyat bolalar kasalxonasi · namuna

## 4 QANDAY ISHLAYDI

### 4 qadam, minimal matn

1. **Qayerga?** Qidiring yoki xaritaga bosing
2. **Variantlar** Eng tez yo‘l va almashinuvlar
3. **Yo‘lga** Avtobus necha daqiqada keladi
4. **Yetib keldingiz** Qolgan bekatlar va vaqt

**RAPTOR** algoritmi telefon ichida 1 soniyadan tez hisoblaydi. Manzilga avtobus bormasa — **eng yaqin bekatgacha** yo‘l va qolgan piyoda masofa ko‘rsatiladi.
**Flutter** (Android · iOS · Web) · MapLibre + OpenStreetMap · GTFS

## 5 BEKAT 2.0 · KEYINGI BOSQICHLAR

namuna

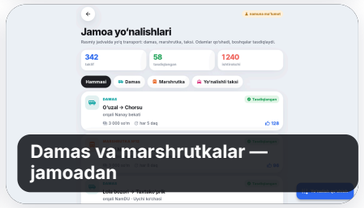
Prototipda allaqachon ishlaydi — namuna ma’lumotlar bilan










## 6 ANALOGLAR

| | Yandex | OSM ilovalar | Bekat |
|---|---|---|---|
| Jonli avtobus | Bor | Yo‘q | 2-bosqich |
| Tarmoq sxemasi | Yo‘q | Yo‘q | Bor |
| Eng yaqin nuqta | — | — | Bor |
| Talab tahlili | Yo‘q | Yo‘q | Bor |
| Namangan bazasi | Yopiq | 0 | Ochiq |

## 7 REJA · SMETA · KPI

| | | |
|---|---|---|
| **40 mln** so‘m smeta | **3 oy** pilotgacha | **Bepul** aholi uchun · B2G |

1–3 h GPS bilan bekatlar · 3–5 h Ilova, 3 til · 6–8 h Marshrut, sxema, reliz · 9–12 h 100+ foydalanuvchi pilot · Keyin Jonli GPS, Andijon

| | | | |
|---|---|---|---|
| **100%** yo‘nalishlar raqamli | **< 1 s** marshrut | **100+** pilot foydalanuvchi | **80%+** yo‘lni topdi |


**Qodirov Ulug‘bek · Zamalitdinov Salim · NamDTU** +998 90 555 80 91 · salzamov@gmail.com · github.com/ZedRubilan/bekat


Xarita © OpenStreetMap, OpenFreeMap · №5, 7, 9 — namuna